ANALYSIS REPORT

Phase 3 Quasi-Static Compliance PINN Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes Wave 5.2 Phase 3 of the sixteen-phase full-PINN theory-validation roadmap. Phase 3 implemented and tested bounded quasi-static compliance, temperature-conditioned stiffness, direction-specific nonlinear compliance, hard elastic-offset equations, and shared-stiffness formulations.

The canonical campaign and its bounded stability follow-up completed:

- main campaign: `phase3_quasi_static_compliance_pinn_2026_07_26`;
- stability campaign: `phase3_c1_fw_stability_repeat_2026_07_26`;
- dataset: `polished_dataset`, causal `setpoints` input mode;
- directional split: `675` train, `194` validation, and `97` test curves;
- main campaign: `12 / 12` completed, `0` failed;
- stability campaign: `2 / 2` completed, `0` failed;
- evaluated surfaces: separate `Fw`, `Bw`, and bounded `global`;
- Phase 3 physical-ingredient promotion: none;
- accepted-reference replacement: none.

  Phase 3 is complete as a valid negative result. The tested compliance laws are implemented, differentiable, inspectable, and physically bounded, but no formulation passed the complete predictive-stability exit gate.

Evidence Contract

Every run reused the immutable common-split signature `c1aa8718fb9bf88cc2021c121dc4f3b4010fc1d2e45ac90af5f4376aa64f8e16`.

Three anomalous training conditions remained quarantined. Directional arms used `675 / 194 / 97` curves and global controls used `1,350 / 388 / 194`. Every candidate preserved:

- causal angle, speed, nominal torque, temperature, and direction inputs;
- the audited convention that measured torque is negative for `Fw` and positive for `Bw`;
- stride `8`, batch size `4`, and a `4,096`-point curve cap;
- the same nine periodic orders;
- a `20`-epoch ceiling and patience `5`;
- target-free physical residuals;
- separate raw, offset, centered-shape, peak-to-peak, harmonic-amplitude, and harmonic-phase evidence. The stability follow-up changed only the random initialization contract. It added `training.random_seed` values `314159` and `271828` and called `seed_everything(seed, workers=True)` before model and dataloader creation.

Candidate Definitions

Arm	Mean Or Physical Law	Scientific Role
`C0`	learned mean plus zero-mean Fourier branch	matched data-driven control; not a PINN
`C1`	learned mean with bounded linear-compliance derivative residual	isolated soft compliance PINN
`C2`	`C1` with temperature-conditioned bounded stiffness	temperature-transfer PINN
`C3`	`C1` with bounded odd nonlinear torque response	nonlinear monotonic-compliance PINN
`C4`	hard direction-specific elastic equation plus periodic residual	hard white-box offset PINN
`C5`	hard shared stiffness with direction-specific intercepts	paired-direction compatibility PINN

For `C1`, the target-free differential residual is `d mean_TE / d tau_signed = 1 / k_direction`.

All stiffness values are strictly bounded through `k = k_min + (k_max - k_min) * sigmoid(raw_k)`.

`C4` and `C5` place the elastic equation directly in the forward path: `mean_TE = intercept_direction + tau_signed / k`.

The learned periodic branch contains sine and cosine terms only and therefore has zero continuous-cycle mean by construction.

Identifiability And Implementation Evidence

The training-only audit passed before campaign preparation:

- `1,932` eligible directional curves;
- full-rank `C1`, `C2`, `C3`, and `C5` design matrices;
- training-only stiffness estimates near `27.2` to `27.5 kNm/deg`;
- positive unconstrained compliance slopes;
- no evidence for ordered load-unload loops or direction-reversal state.

  That last point is important: Phase 3 can test quasi-static elastic compliance, but it cannot claim hysteresis, friction memory, or backlash-state identification from static condition files. The deterministic model validator proved:

- exact signed-torque reconstruction;
- positive bounded stiffness for all physical formulations;
- finite C0-C5 outputs and losses;
- nonzero gradients through `C1`, `C2`, and `C3` residuals;
- exact hard-equation reconstruction for `C4` and `C5`;
- positive temperature-conditioned compliance;
- nonnegative nonlinear compliance increments;
- shared `C5` stiffness;
- near-zero periodic mean;
- Lightning training and inference-context compatibility.

Scalar Campaign Results

The scalar leaderboard uses the training datamodule's reduced angular sampling. It is a campaign diagnostic, not a promotion decision.

Rank	Arm	Surface	Test MAE [deg]	Test RMSE [deg]	Val MAE [deg]
1	`C1`	`Fw`	0.001495	0.001887	0.001702
2	`C2`	`Fw`	0.001551	0.001950	0.001672
3	`C0`	`Fw`	0.001611	0.002017	0.001774
4	`C2`	`Bw`	0.001624	0.002068	0.001727
5	`C3`	`Fw`	0.001745	0.002209	0.001942
6	`C0`	`Bw`	0.001825	0.002313	0.001927
7	`C1`	`Bw`	0.001877	0.002386	0.001970
8	`C3`	`Bw`	0.001926	0.002441	0.002038
9	`C0`	`global`	0.002050	0.002529	0.001977
10	`C4`	`Fw`	0.002087	0.002481	0.002301
11	`C5`	`global`	0.002103	0.002550	0.002448
12	`C4`	`Bw`	0.002350	0.002859	0.002758

The scalar screen selected `C1-Fw` for the authorized stability gate. It did not promote the arm.

Full-Curve Multi-Index Screen

The bounded CVP 1.2 replay evaluated the twelve Phase 3 checkpoints and four accepted references at full source-curve resolution. This is closeout evidence, not the heavy official TE Curve Verification Pipeline refresh.

Forward Surface

Candidate	Raw MAE	Abs. Offset	Centered MAE	P2P [%]	Amp. Err. [%]	Phase [deg]
accepted periodic GRU	0.001618	0.000690	0.001382	8.445	19.237	12.167
accepted periodic MLP	0.001694	0.000825	0.001390	11.802	15.380	13.592
`C1`	0.001843	0.000932	0.001481	10.009	34.128	21.434
`C2`	0.001910	0.000979	0.001563	6.550	49.504	23.278
`C0`	0.001953	0.000989	0.001608	6.463	40.930	22.373
`C3`	0.002068	0.001099	0.001662	11.284	44.033	32.643
`C4`	0.002387	0.001585	0.001476	8.996	30.879	18.527

`C1-Fw` improved over `C0-Fw` on raw error, absolute offset, centered shape, harmonic amplitude, and phase. It worsened peak-to-peak error and remained behind both accepted references. This qualified it for repeat seeds, not for acceptance.

Backward Surface

Candidate	Raw MAE	Abs. Offset	Centered MAE	P2P [%]	Amp. Err. [%]	Phase [deg]
accepted periodic GRU	0.001837	0.000606	0.001649	6.641	31.498	15.321
accepted periodic MLP	0.001912	0.000612	0.001677	11.625	21.132	17.127
`C2`	0.002097	0.000746	0.001859	6.424	47.274	30.608
`C0`	0.002283	0.001068	0.001830	6.108	39.341	28.223
`C1`	0.002331	0.000973	0.001985	5.992	69.396	33.184
`C3`	0.002375	0.001051	0.001978	7.398	60.411	25.984
`C4`	0.002813	0.001655	0.002031	8.019	58.130	32.574

`C2-Bw` improved raw and offset error over `C0-Bw`, but centered shape, harmonic amplitude, and phase regressed. No backward arm passed the joint gate.

Global Surface

Candidate	Raw MAE	Abs. Offset	Centered MAE	P2P [%]	Amp. Err. [%]	Phase [deg]
`C0-global`	0.002386	0.001238	0.001834	7.977	51.201	42.068
`C5-global`	0.002507	0.001635	0.001643	8.577	39.428	23.656

The shared-stiffness equation improved centered shape and harmonic fidelity, but regressed raw error and offset. It therefore failed the global gate.

C1-Fw Initialization-Stability Audit

The initial C1-Fw run and two seeded repeats were replayed on the same full-resolution `97` -curve Fw test surface. Stiffness is in `Nm/deg` ; raw, offset, and centered errors are in degrees; amplitude is in percent; phase is in degrees.

Run	Seed	Stiffness	Raw	Offset	Centered	Amp.	Phase	Gate
initial C1-Fw	not recorded	28,275.59	0.001843	0.000932	0.001481	34.128	21.434	pass
C1-Fw repeat	314159	27,259.39	0.001816	0.000872	0.001513	41.194	20.030	pass
C1-Fw repeat	271828	28,958.11	0.002198	0.001350	0.001605	45.693	22.966	fail

The physical parameter itself is stable

- mean fitted forward stiffness: `28,164.36 Nm/deg` ;
- population standard deviation: `697.95 Nm/deg` ;
- population coefficient of variation: `2.48%` ;
- stiffness-bound loss: zero for all three checkpoints;
- compliance-monotonicity loss: zero for all three checkpoints.

  The predictive benefit is not stable. The `271828` repeat worsens raw error, absolute offset, and harmonic-amplitude fidelity relative to `C0-Fw` . Consequently, only `2 / 3` runs pass the predeclared curve-first gate.

Exit-Gate Decision

The Phase 3 exit rule was

The rule requires stiffness-like parameters to be stable, physically signed, and predictive outside the fitting conditions.

Decision: **Phase 3 complete; no compliance residual promoted.**

Rationale

1. the positive bounded stiffness estimate is numerically stable, so the parameterization itself works;
2. the initial C1-Fw signal is reproducible for one seed but not for both authorized repeats;
3. C2-Bw and C5-global trade raw or offset error against shape and harmonic improvements instead of improving the full surface;
4. hard C4 equations underfit raw and offset behavior;
5. all Phase 3 candidates remain behind the accepted periodic GRU and periodic harmonic MLP on their valid surfaces;
6. the dataset lacks ordered reversal cycles required to reinterpret static offsets as hysteresis or backlash state. No Phase 3 loss weight becomes a default ingredient for later compositions. The implementation and audits remain reusable falsification evidence.

Registry And Program Status

- The scalar main-campaign winner is `C1-Fw`.
- The scalar stability-campaign winner is C1-Fw seed `314159`.
- Neither scalar winner is promoted.
- The current program-best registry remains `te_periodic_gru_sequence_bw`.
- The accepted Fw time-windowed and non-windowed references remain unchanged.
- No official TE Curve Verification Pipeline recommendation changes.
- Wave 5.2 advances to Phase 4 hysteresis, friction, and memory feasibility.

Retained Engineering Value

Phase 3 delivers

- an inspectable C0-C5 quasi-static compliance model family;
- explicit signed-torque reconstruction;
- bounded stiffness and direction intercept parameters;
- soft differential residuals and hard elastic equations;
- temperature-conditioned and nonlinear-compliance ablations;
- separate periodic and quasi-static outputs;
- reproducible model and worker seeding for future campaigns;
- checkpoint-level physical-parameter extraction;
- full-curve raw, offset, centered, peak-to-peak, and harmonic diagnostics;
- local and remote campaign launchers;
- TwinCAT-oriented intermediate quantities.

  These components remain available as controls, diagnostic terms, or future reformulation ingredients. They are not validated physical ingredients at the tested weights and architecture.

Artifact Inventory

Main campaign

- `output/training_campaigns/2026-07-26-17-46-18_phase3_quasi_static_compliance_pinn_2026_07_26/` ;
- `campaign_leaderboard.yaml` ;
- `campaign_best_run.yaml` ;
- `campaign_best_run.md` ;
- `candidate_registries/` .

Stability campaign:

- `output/training_campaigns/2026-07-26-19-52-45_phase3_c1_fw_stability_repeat_2026_07_26/` ;
- two immutable seeded training runs;
- repeat candidate-registry snapshots.

Curve-first evidence:

- `output/validation_checks/phase3_quasi_static_compliance_pinn_curve_payload_diagnostics/2026-07-26-19-39-21__track2c_curve_payload_diagnostics/` ;
- `output/validation_checks/phase3_c1_fw_stability_curve_payload_diagnostics/2026-07-26-20-10-35__track2c_curve_payload_diagnostics/` ;
- `output/analysis/pinn_program_compliance/phase3_c1_fw_stability_audit.yaml` ;
- `output/analysis/pinn_program_compliance/phase3_c1_fw_stability_audit.csv` .

Documentation:

- Phase 3 technical document and campaign plan;
- compliance identifiability audit;
- quasi-static compliance PINN model report;
- C1-Fw stability audit;
- main and stability launcher notes;
- this Markdown report and its validated styled PDF companion.

Next Step

Advance to Phase 4 as a feasibility-first gate:

1. audit whether source files preserve ordered acquisition, reversals,
   repeated cycles, and causal warm-up state;
2. distinguish directly trainable hysteresis laws from synthetic or
   offline-oracle-only tests;
3. do not launch Bouc-Wen, rolling-friction, play/stop, or stateful training
   unless repeated reversal cycles and stable causal state evolution are demonstrably available;
4. preserve a matched NARX or GRU causal-history comparator;
5. keep all rejected Phase 2 and Phase 3 physics weights at zero by default.